

SECSR2043 OPERATING SYSTEMS

[20 Marks]

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Section : 02

Marks

Instruction: Please answer all of the following questions. Whenever the 🙋 symbol appears, please raise your hand to call your instructor, he/she will verify your results by putting his / her initial next to the symbol.

1. Type the following commands using a text editor and save it as a *yourname.sh* (Example: *ahmad.sh*).

```
echo "Hello world" > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo "Honda Accord" > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates;
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt >
orange.txt
cd drinks; cp ../cars/*. *; cp ../fruits/*. *;
cp ../*.jar .
```

- a) Execute the script and draw a tree structure that contains created directories and files. The parent node of the directory begin with **\$HOME** directory.

[4 marks]



Print screen the script that you type;

```

GNU nano 7.2 fathi.sh *
#!/bin/bash

echo "Hello world" > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo "Honda Accord" > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates; date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt > orange.txt
cd ../drinks; cp ../cars/*.c .; cp ../fruits/*.c .; cp ../*.jar .

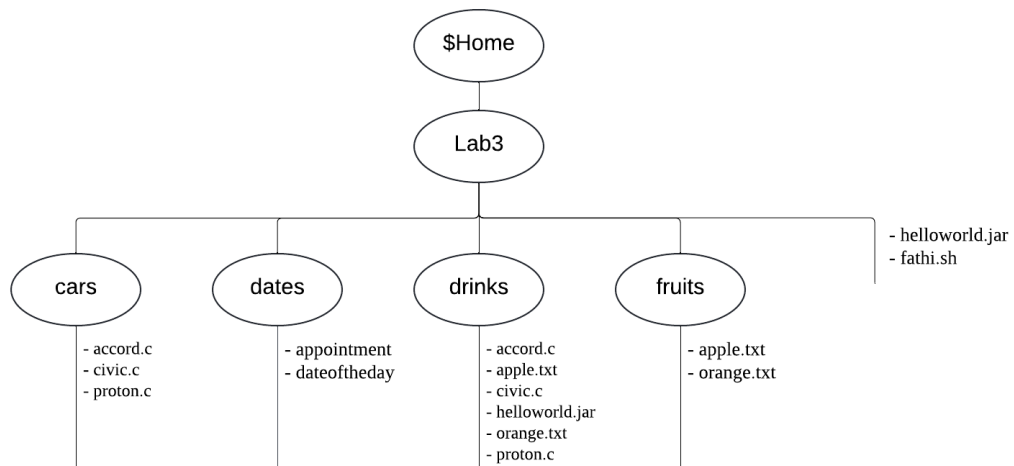
```

```

fathi@secre2043:~/Lab3$ nano fathi.sh
fathi@secre2043:~/Lab3$ chmod u+x fathi.sh
fathi@secre2043:~/Lab3$ ./fathi.sh
fathi@secre2043:~/Lab3$ |

```

Then draw the tree



- b) Write an interactive bash script that will read a type of file extension, display all those files, and count the number of files. To validate your script, display c program files, and enter "c" as the input to the bash script. [4 marks]

Print screen the bash script you type and run

```
GNU nano 7.2 ifiles.sh *
#!/bin/bash

read -p "Enter the file extension (without dot): " ext

echo "Files with ${ext} extension:"
find . -type f -name "*${ext}"

count=$(find . -type f -name "*${ext}" | wc -l)
echo "Number of files with ${ext} extension: $count"
```

```
fathi@secr2043:~/Lab3$ nano ifiles.sh
fathi@secr2043:~/Lab3$ chmod u+x ifiles.sh
fathi@secr2043:~/Lab3$ ./ifiles.sh
Enter the file extension (without dot): c
Files with .c extension:
./drinks/civic.c
./drinks/accord.c
./drinks/proton.c
./cars/civic.c
./cars/accord.c
./cars/proton.c
Number of files with .c extension: 6
fathi@secr2043:~/Lab3$ |
```



2. The following Figure 1 illustrates a tree structure of some directories and files.

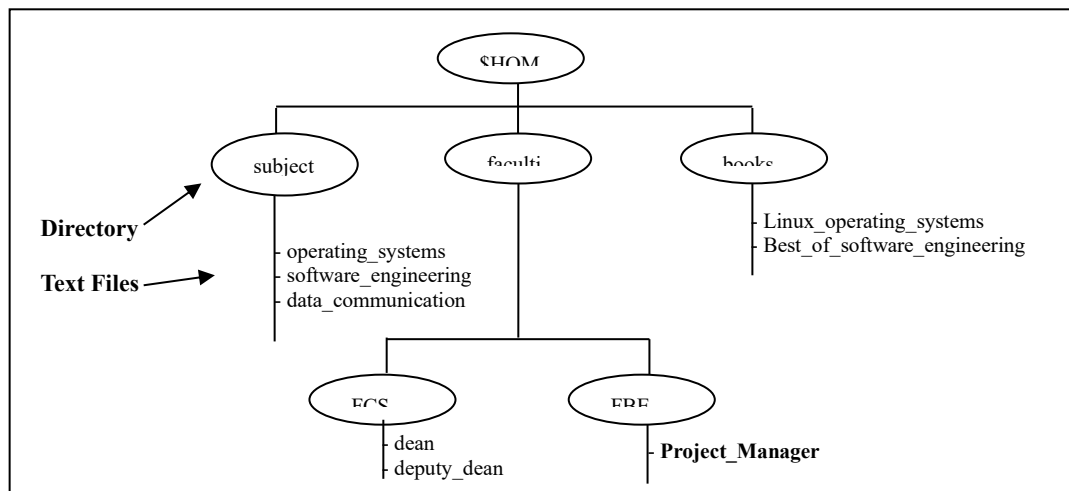


Figure 1

- a) Write a bash script (called `myname2a.sh`) that will produce directories and files as in Figure 1. Each text files contain its filename without the underscore character. For example: text file `Project_Manager` contains `Project Manager`). [4 marks]

Print screen the bash script you type and run

```

GNU nano 7.2 fathi2a.sh *
#!/bin/bash

mkdir -p $HOME/subjects
mkdir -p $HOME/faculties/FCS
mkdir -p $HOME/faculties/FBE
mkdir -p $HOME/books

echo "operating systems" > $HOME/subjects/operating_systems
echo "software engineering" > $HOME/subjects/software_engineering
echo "data communication" > $HOME/subjects/data_communication

echo "dean" > $HOME/faculties/FCS/dean
echo "deputy dean" > $HOME/faculties/FCS/deputy_dean

echo "Project Manager" > $HOME/faculties/FBE/Project_Manager

echo "Linux operating systems" > $HOME/books/Linux_operating_systems
echo "Best of software engineering" > $HOME/books/Best_of_software_engineering

fathi@secr2043:~$ nano fathi2a.sh
fathi@secr2043:~$ chmod u+x fathi2a.sh
fathi@secr2043:~$ ./fathi2a.sh
  
```



- b) Complete the following table by writing the access control of directories or files that were produced. Given is the access control for directory called book.

[2 marks]

Directory/File	Access Control
books	drwxrwxr-x
subjects	drwxrwxr-x
Best_of_software_engineering	-rw-rw-r--
FCS	drwxrwxr-x
project_manager	-rw-rw-r--



```
drwxrwxr-x 2 fathi fathi 4096 Jun 27 11:43 books
```

```
drwxrwxr-x 2 fathi fathi 4096 Jun 27 11:43 subjects
```

```
drwxrwxr-x 2 fathi fathi 4096 Jun 27 11:43 FCS
```

```
-rw-rw-r-- 1 fathi fathi 16 Jun 27 11:43 Project_Manager
```

```
-rw-rw-r-- 1 fathi fathi 29 Jun 27 11:43 Best_of_software_engineering
```

- c) Write another bash script (called `myname2c.sh`) that will change the access control of the directories and files based on the following information:

[4 marks]

Directory/File	Users								
	Owner			Group			Public		
subjects	✓	✓	✓	✓	x	x	✓	x	x
Best_of_software_engineering	✓	x	✓	x	✓	x	x	x	x
FCS	✓	✓	x	x	x	x	✓	✓	✓
project_manager	x	x	x	x	✓	✓	x	x	✓

Print screen the bash script you type and run

```

GNU nano 7.2                                fathi2c.sh *
#!/bin/bash

chmod 744 $HOME/subjects
chmod 520 $HOME/books/Best_of_software_engineering
chmod 607 $HOME/faculties/FCS
chmod 031 $HOME/faculties/FBE/Project_Manager

fathi@secr2043:~$ nano fathi2c.sh
fathi@secr2043:~$ chmod u+x fathi2c.sh
fathi@secr2043:~$ ./fathi2c.sh

drwxr--r-- 2 fathi fathi 4096 Jun 27 11:43 subjects
-r-x-w---- 1 fathi fathi 29 Jun 27 11:43 Best_of_software_engineering
drw----rwx 2 fathi fathi 4096 Jun 27 11:43 FCS
-----wx--x 1 fathi fathi 16 Jun 27 11:43 Project_Manager

```



- d) Complete the following table by writing the access control for each directory or file after executing the bash script in question 2(c). [2 marks]

Directory/File	Access Control
subjects	drwxr--r--
Best_of_software_engineering	-r-x-w----
FCS	drw----rwx
project_manager	-----wx--x

End of Lab 3

*** *All the Best for Final Exam* ***